Yinsheng Yao

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EDUCATION

Tongji University Shanghai, China
Bachelor of Engineering in Computer Science and Technology *Sep. 2022 – Jun. 2026 (Expected)*
GPA: 88.81/100
Selected Honors: Second Prize, Tongji University Scholarship (2022-2023 Academic Year)

RESEARCH EXPERIENCE

Computational Art History Benchmark for Classical Calligraphy Criticism *Mar. 2026 – Present*
Group Project, Supervisor: Prof. Chen Ye, Tongji University

- Constructed the first cross-modal benchmark mapping historical Chinese calligraphy criticism texts (Tang–Qing dynasties) to visual style discrimination tasks, bridging computational vision and digital humanities.
- Designed zero-shot/few-shot style identification experiments using frontier MLLMs (GPT-4o, Qwen-VL), quantifying how classical rhetorical schools (e.g., Kang Youwei's Bei school vs. Tang-style imagery) guide AI visual grounding.
- Introduced controlled ablations (classical Chinese vs. modern paraphrase prompts) to isolate cross-modal decoding ability from language comprehension bias, revealing cognitive blind spots in MLLMs on abstract aesthetic metaphors.

Data-Model Coupled Quality Recognition and Optimization for Power Industry *Feb. 2026 – Present*
Group Project, Supervisor: Prof. Dawei Cheng, Tongji University

- Responsible for Research Module 3: quality-driven training optimization strategies for power domain language models.
- Designed a quantitative evaluation mechanism for sample quality impact on model performance, establishing mathematical mappings between quality indicators and model capability across fault diagnosis, intelligent Q&A, and operation assistance scenarios.
- Developed a quality-aware training strategy combining active learning and curriculum learning, enabling adaptive adjustment of training schedules based on sample quality for progressive model optimization.
- Prototyped a sample quality evaluation toolkit to support continuous co-optimization of data quality and model cognition.

Hierarchical Benchmark for Automated Financial Research Reporting *Oct. 2025 – Feb. 2026*
Group Project, Supervisor: Prof. Dawei Cheng, Tongji University

- Designed and constructed a benchmark for deep analysis of financial research reports, targeting causal reasoning, factual grounding, risk identification, and comparative analysis in long-form financial texts.
- Led the development of the financial deep analysis module, defining fine-grained evaluation dimensions and building a structured QA dataset with rule-based and numerical reasoning challenges.
- Contributed to a research paper based on this benchmark; paper targeting submission to NIPS 2026.

Graph-LLM for Interbank Risk-Aware Topological Inference and Credit Rating *Nov. 2025 – Feb. 2026*
Group Project, Supervisor: Prof. Dawei Cheng, Tongji University

- Co-authored GLM4Risk (2nd author), a Graph-LLM framework aligning GNN structural embeddings with LLM representations for interbank credit rating and risk-aware topological inference.
- Responsible for reproducing and evaluating state-of-the-art Graph-LLM baselines (GraphGPT, LLaGA, ConGraT, etc.) on a real-world dataset of 4,548 global banks over 32 quarters, and contributed to manuscript writing.
- Contributed to writing the paper, which was submitted to KDD 2026.

Multi-Agent Reasoning and Evaluation Framework *Aug. 2025 – Jan. 2026*
Group Project, Supervisor: Prof. Dawei Cheng, Tongji University

- Designed and implemented a multi-agent reasoning architecture integrating specialized roles (Planner, Researcher, Skeptic, Judge, and Stylist) to support multi-stage planning, internal debate, external verification, and consensus synthesis for complex reasoning tasks.
- Developed a reasoning-hop-based routing mechanism, enabling dynamic path selection between fast and full reasoning pipelines based on task complexity; Implemented asynchronous web verification and structured deliberation loops to improve factual consistency and causal reasoning.
- Conducted automated evaluations on OBQA, TruthfulQA, HALUEVAL and StrategyQA datasets, achieving great performance compared with the baseline.
- Contributed to writing a research paper based on this framework as the first author; paper targeting submission to NIPS 2026.

Detect and Correct: Hybrid Feedback Framework for Code Hallucination Mitigation *May 2025 – Present*
Group Project, Supervisor: Prof. Tianyi Zhang, Purdue University

- Designed a hybrid feedback framework combining model-based hallucination detection and execution-based verification to reduce logical errors in LLM-generated code.
- Built a Transformer detector using token, syntax, and confidence features to localize logical errors.
- Designed an iterative correction pipeline with structured hybrid prompts and a "reset thinking" mechanism.

- Improved absolute code accuracy by 9-20% compared to greedy decoding on HumanEval, MBPP, LiveCodeBench, and BigCodeBench benchmarks.
- Currently contributing to manuscript writing as the first author; paper targeting submission to EMNLP 2026.

Chinese Character Style Recognition System *Jan. 2023 – Mar. 2026*

Group Project, Supervisor: Prof. Chen Ye, Tongji University

- Built a deep learning framework for 400 calligraphy styles (1,000+ images each) using Vision Transformer (ViT) with pretrained ImageNet weights via timm.
- Enhanced discrimination through ArcFace loss function, improving the separation of visually similar calligraphic styles
- Applied TrivialAugmentWide, Mixup, CutMix, and optimized training with AdamW, cosine annealing, AMP, and early stopping for efficient, stable convergence.
- Developed an automated preprocessing pipeline (SVM-based routing, adaptive binarization, CCA denoising) to standardize large-scale calligraphy data.
- Constructed a benchmark comparing CNN and ViT architectures under simple vs. strong augmentation, providing reproducible baselines for style recognition.
- Built HCSU, the first fine-grained calligraphy dataset with 39,307 images from 49 calligraphers across 10 dynasties, decoupling ink manuscripts (Tie) from stone rubbings (Bei) to resolve modal mixture; provided hierarchical aesthetic descriptions for style discrimination and aesthetic reasoning evaluation.
- Authored a paper (ECCV 2026, under review, first author) revealing a knowledgeable but unperceptive paradox in state-of-the-art LVLMs on cultural heritage visual tasks.

Other Project Deliverables and Achievements:

- Developed and deployed three web applications: a project showcase, a calligraphy style recognition system & showcase, and a character image retrieval site, built with HTML, CSS, and JavaScript.
- Submitted a deployed system as a project and received the Excellent Project Award in the University Innovation and Entrepreneurship Training Program.
- Presented project at the "Between the Lines: The Beauty of Chinese Characters and the Path of Intelligent Evolution" exhibition at Tongji University Museum.
- Relevant work was accepted by the 1st CCF AI for Humanities Conference (Beijing, China, 2025); the dataset was selected by the conference as a recommended dataset case (Top 40).

SELECTED PROJECTS

Multilingual Sign Language Processing System Based on Large Language Models *Dec. 2024 – Oct. 2025*

Group Project (Team Leader)

- Designed word order transformation logic between natural and multilingual sign languages for video synthesis
- Enhanced gesture precision by comparing digital human motions with teacher demonstrations
- Structured sign sequences in JSON for integration with video generation pipelines
- Validated gesture consistency and created video batches for the National Innovation and Entrepreneurship Program
- Awarded in the 2025 China International College Students' Innovation Competition (Shanghai Division)

Novel Style Discovery & Dataset Construction Module *Mar. 2024 – Apr 2025*

Group Project, Supervisor: Prof. Chen Ye, National-level Innovation and Entrepreneurship Training Program

- Collected historical Chinese calligraphy data and applied character-level segmentation tools to build a comprehensive style-specific dataset, and built a website with timeline visualizations of calligraphy heritage
- Assisted in training a neural network for writing style discovery using SupConLoss and DINO, and performed PCA-based feature visualization

INTERNSHIP EXPERIENCE

Shanghai Bright Power Semiconductor Co., Ltd. *Jan. – Feb. 2024*

Assistant Test Engineering Intern

- Participated in chip testing, including PCB soldering, electrical connections, and parameter measurement
- Developed C-based testing programs integrated with the CTA8280 system for multiple chip batches
- Analyzed chip performance data on yield and tolerance under extreme voltage/current conditions to support reliability evaluation

PROFESSIONAL SKILL

Programming: Python, C, C++ (Proficient in syntax and object-oriented design), MATLAB, Verilog, Assembly
Frameworks & Tools: PyTorch, timm, Docker, GitHub, Hugging Face
Database & Back-end: MySQL (with basic optimization), OceanBase, MiniOB
Front-end Development: HTML5, CSS, JavaScript
AI Application: Deep Learning, Computer Vision, Transformer Models, Large Language Models (LLMs), Prompt Engineering